

Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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1 Pipeline Validation on MACS-18

1.1 Execution Success Rate

Among the executed data samples, which ones of them completed the entire pipeline without any exception or error.

$$ESR_{Total} = \frac{\text{Number of correctly executed samples}}{\text{Number of total executed samples}} = 100\%$$

Normal_1: True	Diseased_1: True
Normal_2: True	Diseased_2: True
Normal_3: True	Diseased_3: True
Normal_4: True	Diseased_4: True
Normal_5: True	Diseased_5: True
Normal_6: True	Diseased_6: True
Normal_7: True	Diseased_7: True
Normal_8: True	Diseased_8: True
Normal_9: True	Diseased_9: True
Normal_10: True	Diseased_10: True
Normal_11: True	Diseased_11: True
Normal_12: True	Diseased_12: True
Normal_13: True	Diseased_13: True
Normal_14: True	Diseased_14: True
Normal_15: True	Diseased_15: True
Normal_16: True	Diseased_16: True
Normal_17: True	Diseased_17: True
Normal_18: True	Diseased_18: True
Normal_19: True	Diseased_19: True
Normal_20: True	Diseased_20: True

1.2 Mean Detected Endpoints per Patient

In comparison with the ASOCA dataset (17.1 endpoints per patient), MACS-18 has more detailed geometrical information so more endpoints are detected on average for each patient, confirming that the improved data gives more details about the geometry of the coronary arteries.

$$MPE_{Normal} = \frac{\text{Number total detected endpoints in Normal samples}}{\text{Number of Normal samples}} = 30.75 \text{ endpoints}$$

$$MPE_{Diseased} = \frac{\text{Number total detected endpoints in Diseased samples}}{\text{Number of Diseased samples}} = 24.65 \text{ endpoints}$$

$$MPE_{Total} = \frac{\text{Number total detected endpoints}}{\text{Number of Total Samples}} = 27.2 \text{ endpoints}$$

Normal_1: 28	Diseased_1: 16
Normal_2: 25	Diseased_2: 24
Normal_3: 42	Diseased_3: 27
Normal_4: 22	Diseased_4: 32
Normal_5: 24	Diseased_5: 33
Normal_6: 22	Diseased_6: 19
Normal_7: 27	Diseased_7: 28
Normal_8: 28	Diseased_8: 21
Normal_9: 35	Diseased_9: 42
Normal_10: 31	Diseased_10: 15
Normal_11: 57	Diseased_11: 20
Normal_12: 17	Diseased_12: 12
Normal_13: 64	Diseased_13: 21
Normal_14: 33	Diseased_14: 23
Normal_15: 30	Diseased_15: 21
Normal_16: 26	Diseased_16: 31
Normal_17: 23	Diseased_17: 23

Normal_18: 25	Diseased_18: 32
Normal_19: 24	Diseased_19: 27
Normal_20: 32	Diseased_20: 26

1.3 Ostium Detection Accuracy

The Ostium Detection Accuracy (ODA) evaluates the anatomical accuracy of the automated ostium detection algorithm. This metric was obtained by visually inspecting the spatial coordinates of the algorithmically detected ostiums and comparing them against the true anatomical origins of the left and right coronary arteries. It represents the ratio of correctly placed ostium points to the total number of ostiums expected (two per patient).

$$OISR_{Normal} = \frac{\text{Number of correctly identified ostium points}}{\text{Number of total ostium points}} = 75.00\%$$

$$OISR_{Diseased} = \frac{\text{Number of correctly identified ostium points}}{\text{Number of total ostium points}} = 72.50\%$$

$$OISR_{Total} = \frac{\text{Number of correctly identified ostium points}}{\text{Number of total ostium points}} = 73.75\%$$

Normal_1: 1/2	Diseased_1: 2/2
Normal_2: 0/2	Diseased_2: 2/2
Normal_3: 2/2	Diseased_3: 1/2
Normal_4: 1/2	Diseased_4: 1/2
Normal_5: 2/2	Diseased_5: 1/2
Normal_6: 1/2	Diseased_6: 2/2
Normal_7: 2/2	Diseased_7: 1/2
Normal_8: 2/2	Diseased_8: 1/2
Normal_9: 2/2	Diseased_9: 1/2
Normal_10: 1/2	Diseased_10: 2/2
Normal_11: 1/2	Diseased_11: 0/2
Normal_12: 2/2	Diseased_12: 2/2
Normal_13: 2/2	Diseased_13: 2/2
Normal_14: 1/2	Diseased_14: 1/2
Normal_15: 1/2	Diseased_15: 2/2
Normal_16: 2/2	Diseased_16: 2/2
Normal_17: 1/2	Diseased_17: 1/2
Normal_18: 2/2	Diseased_18: 2/2
Normal_19: 2/2	Diseased_19: 1/2
Normal_20: 2/2	Diseased_20: 2/2

Discussion: The observed decrease in Ostium Identification Success Rate (OISR) when transitioning from the ASOCA to the MACS-18 dataset can be directly attributed to the higher spatial resolution and topological complexity of the latter. High-resolution CCTA imaging captures a greater degree of anatomical detail, including minor side branches, surface irregularities, and local topological variations. Consequently, the automated 3D processing pipeline generates a significantly higher number of endpoints (averaging 27.7 endpoints per patient in the MACS-18 cohort). This increased topological density introduces algorithmic noise; with a larger pool of potential root candidates.

1.4 Centerline Extraction Precision

$$CESR_{Normal} = \frac{\text{Number of correctly extracted centerlines}}{\text{Number of attempted centerline extractions}} = 96.87\%$$

$$CESR_{Diseased} = \frac{\text{Number of correctly extracted centerlines}}{\text{Number of attempted centerline extractions}} = 94.04\%$$

$$CESR_{Total} = \frac{\text{Number of correctly extracted centerlines}}{\text{Number of attempted centerline extractions}} = 95.62\%$$

Normal_1: 26/26	Diseased_1: 12/14
Normal_2: 23/23	Diseased_2: 20/22
Normal_3: 38/40	Diseased_3: 25/25
Normal_4: 20/20	Diseased_4: 25/30
Normal_5: 22/22	Diseased_5: 30/30

Normal_6: 20/20	Diseased_6: 16/17
Normal_7: 25/25	Diseased_7: 26/26
Normal_8: 24/26	Diseased_8: 19/19
Normal_9: 20/33	Diseased_9: 39/40
Normal_10: 28/29	Diseased_10: 13/13
Normal_11: 54/55	Diseased_11: 17/18
Normal_12: 15/15	Diseased_12: 10/10
Normal_13: 62/62	Diseased_13: 19/19
Normal_14: 30/31	Diseased_14: 7/21
Normal_15: 27/28	Diseased_15: 19/19
Normal_16: 21/24	Diseased_16: 28/29
Normal_17: 21/21	Diseased_17: 21/21
Normal_18: 23/23	Diseased_18: 30/30
Normal_19: 22/22	Diseased_19: 25/25
Normal_20: 26/30	Diseased_20: 24/24

Discussion: Contrary to the challenges observed in ostium identification, the Centerline Extraction Success Rate (CESR) demonstrated a noticeable improvement when processing the MACS-18 dataset. This increase in algorithmic robustness can be directly attributed to the superior spatial resolution and geometric fidelity of the MACS-18 CCTA scans. Centerline extraction algorithms, such as those implemented in VMTK, rely heavily on the internal Voronoi diagram of the 3D surface mesh to compute optimal paths. In lower-resolution datasets like ASOCA, severe stenoses or highly distal vessel segments often suffer from partial volume effects, leading to artificial surface collapsing, mesh discontinuities, or degenerate triangles. These topological artifacts cause the pathfinding algorithm to encounter dead ends or fail to compute the inscribed spheres, resulting in aborted centerlines. Conversely, the high-resolution surface meshes generated from the MACS-18 dataset preserve the continuous lumen topology even across severely diseased and narrowed segments. This continuous geometric definition provides a highly regular and unbroken Voronoi network, allowing the algorithm to successfully trace and connect distal endpoints back to the root, thereby achieving a 95.62\% global extraction success rate.

1.5 Block and Total Runtimes

Mean Total Runtime = 576.18s $\approx$ 9.60 minutes

Mean Block 1 Runtime (3D Anatomy Extraction) = 159.76s

Mean Block 2 Runtime (Geometric Quantification) = 357.73s

Mean Block 3 Runtime (Labeling + CADRADs Scoring) = 57.86s

Mean Block 4 Runtime (Dashboard Setup) = 0.02s

2 Pipeline Validation on ASOCA

1.1 Execution Success Rate

Among the executed data samples, which ones of them completed the entire pipeline without any exception or error.

$$ESR_{Total} = \frac{\text{Number of correctly executed samples}}{\text{Number of total executed samples}} = 100\%$$

Normal_1: True	Diseased_1: True
Normal_2: True	Diseased_2: True
Normal_3: True	Diseased_3: True
Normal_4: True	Diseased_4: True
Normal_5: True	Diseased_5: True
Normal_6: True	Diseased_6: True
Normal_7: True	Diseased_7: True
Normal_8: True	Diseased_8: True
Normal_9: True	Diseased_9: True
Normal_10: True	Diseased_10: True
Normal_11: True	Diseased_11: True
Normal_12: True	Diseased_12: True
Normal_13: True	Diseased_13: True
Normal_14: True	Diseased_14: True

Normal_15: True	Diseased_15: True
Normal_16: True	Diseased_16: True
Normal_17: True	Diseased_17: True
Normal_18: True	Diseased_18: True
Normal_19: True	Diseased_19: True
Normal_20: True	Diseased_20: True

1.2 Endpoint Identification Precision

From those detected endpoints, which are truly correct endpoints.

$$\begin{aligned}
 EISR_{Normal} &= \frac{\text{Number of correctly identified artery endpoints}}{\text{Number of total identified artery endpoints}} = \frac{327}{364} = 89.84\% \\
 EISR_{Diseased} &= \frac{\text{Number of correctly identified artery endpoints}}{\text{Number of total identified artery endpoints}} = \frac{287}{320} = 89.69\% \\
 EISR_{Total} &= \frac{\text{Number of correctly identified artery endpoints}}{\text{Number of total identified artery endpoints}} = \frac{614}{684} = 89.77\%
 \end{aligned}$$

Normal_1: 16/16	Diseased_1: 10/10
Normal_2: 12/12	Diseased_2: 9/11
Normal_3: 16/20	Diseased_3: 14/16
Normal_4: 10/13	Diseased_4: 14/15
Normal_5: 19/20	Diseased_5: 19/21
Normal_6: 11/12	Diseased_6: 17/20
Normal_7: 14/15	Diseased_7: 13/15
Normal_8: 18/19	Diseased_8: 16/18
Normal_9: 13/16	Diseased_9: 15/15
Normal_10: 16/17	Diseased_10: 11/11
Normal_11: 23/23	Diseased_11: 14/16
Normal_12: 14/15	Diseased_12: 14/15
Normal_13: 27/32	Diseased_13: 20/22
Normal_14: 28/30	Diseased_14: 17/19
Normal_15: 10/12	Diseased_15: 10/13
Normal_16: 11/13	Diseased_16: 15/17
Normal_17: 15/16	Diseased_17: 17/19
Normal_18: 19/23	Diseased_18: 19/21
Normal_19: 15/19	Diseased_19: 11/12
Normal_20: 20/21	Diseased_20: 12/14

1.3 Ostium Detection Accuracy

Among the detected ostiums, how many of them were truly correct.

$$\begin{aligned}
 OISR_{Normal} &= \frac{\text{Number of correctly identified ostium points}}{\text{Number of total ostium points}} = \frac{34}{40} = 85\% \\
 OISR_{Diseased} &= \frac{\text{Number of correctly identified ostium points}}{\text{Number of total ostium points}} = \frac{34}{40} = 85\% \\
 OISR_{Total} &= \frac{\text{Number of correctly identified ostium points}}{\text{Number of total ostium points}} = \frac{68}{80} = 85\%
 \end{aligned}$$

Normal_1: 2/2	Diseased_1: 2/2
Normal_2: 2/2	Diseased_2: 2/2
Normal_3: 2/2	Diseased_3: 2/2
Normal_4: 1/2	Diseased_4: 2/2
Normal_5: 2/2	Diseased_5: 1/2
Normal_6: 1/2	Diseased_6: 1/2
Normal_7: 1/2	Diseased_7: 2/2
Normal_8: 1/2	Diseased_8: 2/2
Normal_9: 2/2	Diseased_9: 2/2
Normal_10: 2/2	Diseased_10: 2/2
Normal_11: 1/2	Diseased_11: 2/2
Normal_12: 2/2	Diseased_12: 1/2
Normal_13: 2/2	Diseased_13: 2/2
Normal_14: 2/2	Diseased_14: 1/2

Normal_15: 2/2	Diseased_15: 2/2
Normal_16: 2/2	Diseased_16: 2/2
Normal_17: 2/2	Diseased_17: 1/2
Normal_18: 2/2	Diseased_18: 2/2
Normal_19: 2/2	Diseased_19: 1/2
Normal_20: 1/2	Diseased_20: 2/2

1.4 Centerline Extraction Success Rate

$$CESR_{Normal} = \frac{\text{Number of correctly extracted centerlines}}{\text{Number of attempted centerline extractions}} = \frac{267}{321} = 83.18\%$$

$$CESR_{Diseased} = \frac{\text{Number of correctly extracted centerlines}}{\text{Number of attempted centerline extractions}} = \frac{228}{274} = 83.21\%$$

$$CESR_{Total} = \frac{\text{Number of correctly extracted centerlines}}{\text{Number of attempted centerline extractions}} = \frac{495}{595} = 83.19\%$$

Normal_1: 11/14 (3)	Diseased_1: 8/8 (0)
Normal_2: 10/10 (0)	Diseased_2: 6/8 (2)
Normal_3: 8/18 (10)	Diseased_3: 12/12 (0)
Normal_4: 7/11 (4)	Diseased_4: 11/13 (2)
Normal_5: 18/18 (0)	Diseased_5: 16/19 (3)
Normal_6: 7/10 (3)	Diseased_6: 16/18 (2)
Normal_7: 11/13 (2)	Diseased_7: 11/13 (2)
Normal_8: 15/17 (2)	Diseased_8: 16/16 (0)
Normal_9: 11/14 (3)	Diseased_9: 11/13 (2)
Normal_10: 13/15 (2)	Diseased_10: 8/9 (1)
Normal_11: 17/21 (4)	Diseased_11: 11/14 (3)
Normal_12: 12/13 (1)	Diseased_12: 12/14 (2)
Normal_13: 24/30 (6)	Diseased_13: 15/20 (5)
Normal_14: 27/28 (1)	Diseased_14: 11/17 (6)
Normal_15: 8/10 (2)	Diseased_15: 8/11 (3)
Normal_16: 9/9 (0)	Diseased_16: 9/11 (2)
Normal_17: 12/13 (1)	Diseased_17: 12/17 (5)
Normal_18: 16/21 (5)	Diseased_18: 15/19 (4)
Normal_19: 13/17 (4)	Diseased_19: 9/10 (1)
Normal_20: 18/19 (1)	Diseased_20: 11/12 (1)

1.5 Block and Total Runtimes

Mean Total Runtime = 346.92s $\approx$ 5.78 minutes
Mean Block 1 Runtime (3D Anatomy Extraction) = 145.78s
Mean Block 2 Runtime (Geometric Quantification) = 164.64s
Mean Block 3 Runtime (Labeling + CADRADs Scoring) = 35.19s
Mean Block 4 Runtime (Dashboard Setup) = 0.16s

Normal_1: 139.24s / 149.63s / 34.65s / 0.18s = 325.11s	Diseased_1: 143.48s / 78.80s / 22.35s / 0.14s = 245.45s
Normal_2: 119.24s / 109.53s / 24.81s / 0.16s = 254.39s	Diseased_2: 139.32s / 71.73s / 23.91s / 0.22s = 236.50s
Normal_3: 142.15s / 99.91s / 22.58s / 0.18s = 265.39s	Diseased_3: 149.27s / 142.49s / 39.64s / 0.16s = 332.70s
Normal_4: 135.60s / 82.38s / 20.75s / 0.15s = 239.63s	Diseased_4: 143.79s / 95.05s / 26.78s / 0.16s = 266.84s
Normal_5: 135.23s / 177.30s / 35.25s / 0.14s = 348.77s	Diseased_5: 145.83s / 196.83s / 38.68s / 0.17s = 382.35s
Normal_6: 136.23s / 79.44s / 18.77s / 0.16s = 235.50s	Diseased_6: 159.52s / 186.38s / 38.97s / 0.17s = 385.89s
Normal_7: 132.07s / 144.20s / 30.72s / 0.15s = 308.17s	Diseased_7: 157.96s / 185.73s / 44.54s / 0.18s = 389.33s
Normal_8: 150.06s / 243.35s / 43.43 / 0.16s = 437.82s	Diseased_8: 161.00s / 209.60s / 45.39s / 0.15s = 417.36s
Normal_9: 148.53s / 131.80s / 32.26s / 0.15s = 313.42s	Diseased_9: 153.66s / 167.08s / 38.69s / 0.16s = 360.39s
Normal_10: 147.02s / 204.02s / 39.81s / 0.15s = 391.80s	Diseased_10: 149.62s / 81.41s / 25.54s / 0.16s = 257.60s
Normal_11: 154.40s / 285.19s / 46.85s / 0.13s = 487.58s	Diseased_11: 151.53s / 119.40s / 30.68s / 0.23s = 302.60s
Normal_12: 138.00s / 133.80s / 29.67s / 0.14s = 302.60s	Diseased_12: 135.26s / 101.44s / 30.53s / 0.13s = 268.05s
Normal_13: 152.83s / 386.28s / 57.77s / 0.15s = 597.97s	Diseased_13: 158.82s / 246.72s / 42.33s / 0.18s = 449.00s
Normal_14: 161.72s / 401.24s / 69.84s / 0.13s = 633.60s	Diseased_14: 145.40s / 117.74s / 21.89s / 0.16s = 296.06s
Normal_15: 132.70s / 80.50s / 23.70s / 0.16s = 237.67s	Diseased_15: 155.50s / 119.26s / 34.18s / 0.15s = 309.84s
Normal_16: 138.92s / 126.46s / 30.53s / 0.16s = 297.22s	Diseased_16: 142.84s / 188.57s / 40.60s / 0.19s = 373.33s

Normal_17: 147.71s / 143.04s / 35.33s / 0.17s = 327.33s
Normal_18: 125.10s / 303.59s / 48.89s / 0.16s = 478.68s
Normal_19: 145.21s / 122.97s / 30.35s / 0.16s = 299.43s
Normal_20: 151.84s / 174.30s / 38.29s / 0.16s = 365.33s

Diseased_17: 156.23s / 182.62s / 37.08s / 0.16s = 377.02s
Diseased_18: 165.63s / 233.00s / 45.95s / 0.18s = 445.65s
Diseased_19: 117.79s / 117.38s / 30.73s / 0.16s = 266.81s
Diseased_20: 165.12s / 165.42s / 34.76s / 0.24s = 366.78s

3 Geometric and Algorithmic Verification: Synthetic Data

3.1 Control Model Validation: Synthetic 1 - Healthy

- *Theoretical Maximum %AS* = 0.00%
- *Predicted Maximum %AS* = 2.64%
- *Absolute Error %AS* = $| \textit{Theoretical Max \%AS} - \textit{Predicted Max \%AS} | = 2.64\%$
- *Theoretical Area* = 314.16mm^2
- *Predicted Area Mean* = 319.39mm^2
- *Mean Area Absolute Error* = $| \textit{Theoretical Area} - \textit{Predicted Area Mean} | = 5.23\text{mm}^2$
- *Maximum Area Deviation* = 39.17mm^2

3.2 Stenosis Model Validation: Synthetic 2 - Diseased

- *Theoretical Maximum %AS* = 75.00%
- *Predicted Maximum %AS* = 73.67%
- *Absolute Error Maximum %AS* = $| \textit{Theoretical Max \%AS} - \textit{Predicted Max \%AS} | = 1.33\%$
- *Theoretical Maximum Area* = 314.16mm^2
- *Theoretical Minimum Area* = 78.54mm^2
- *Predicted Maximum Area* = 333.62mm^2
- *Predicted Minimum Area* = 82.55mm^2
- *Absolute Error Maximum Area* = $| \textit{Theoretical Max Area} - \textit{Predicted Max Area} | = 19.47\text{mm}^2$
- *Absolute Error Minimum Area* = $| \textit{Theoretical Min Area} - \textit{Predicted Min Area} | = 4.01\text{mm}^2$

Llindars (Thresholds): En l'àmbit del processament d'imatge mèdica i hemodinàmica computacional, un error absolut de només un **1.33%** en la predicció del percentatge d'estenosi és un èxit rotund. La majoria de variabilitat inter-observador (quan dos metges miren la mateixa imatge manualment) sol estar entre el 5% i el 10%. Per tant, pots defensar que el teu algoritme té una precisió totalment clínica.

Com millorar-ho: Per justificar aquells **5.23 mm²** d'error absolut a l'àrea, pots suggerir en el text que futures versions poden utilitzar mètodes interpolació espacial sub-vòxel encara més fins, o augmentar la densitat dels punts de mostreig per sobre dels 400 talls actuals del *fallback*.

4 User Experience and Interface Usability Evaluation

Results after the users complete the interaction and the formulary:

- <https://docs.google.com/forms/d/e/1FAIpQLSeHmCs-B8pYm0gK3JLsp3BxIJqsrcMEDv3arWlycSDIPj6ARA/viewform?usp=publish-editor>

4.1 System Usability Score

+++

4.2 Feedback, Discussions and Future Work with Doctors at HSP

During the final stages of this project, a clinical evaluation session was conducted with expert cardiologists at Hospital de la Santa Creu i Sant Pau to assess the pipeline's clinical utility and identify areas for future improvement. The feedback gathered is highly valuable, as it bridges the gap between technical implementation and real-world clinical workflows. The following sections outline the key discussions and proposed future work.

4.2.1 Algorithmic and Metric Refinements

- **Dynamic Reference Values for Stenosis Calculation:** The current formula uses fixed cross-sectional area reference values depending on a predefined window size, which introduces limitations in quantification, particularly near bifurcations. Clinical experts noted that they typically select healthy reference points within the same segment manually. Future iterations should allow clinicians to dynamically adjust the proximal and distal reference values, bypassing anatomical anomalies and improving the accuracy of the stenosis calculation.
- **Centerline Resolution Optimization:** The current pipeline extracts centerline points with high spatial granularity (< 0.1 mm for short branches, < 0.2 mm for long branches). While this provides rich geometric data, cardiologists suggested that a coarser spacing (e.g., ~ 2 mm) is generally sufficient for standard diagnosis. Reducing the resolution would significantly improve computational efficiency, though higher granularity could be dynamically retained exclusively in diseased segments.
- **Plaque vs. Stenosis Quantification:** A critical clinical distinction was emphasized during the evaluation: while plaque accumulation (e.g., calcium, cholesterol) causes stenosis, the presence of plaque does not always imply a hemodynamically significant stenosis. Current clinical workflows prioritize identifying plaque first. Future work should integrate plaque quantification alongside stenosis detection, providing a more comprehensive and realistic diagnostic picture.

4.2.2 Visualization and Interface Enhancements

- **Continuous Lumen Visualization:** Clinicians expressed the need to visualize the lumen's progression continuously along the artery, rather than relying solely on discrete centerline data points. Providing continuous cross-sectional profiles allows for an easier and more intuitive diagnostic assessment.
- **Integration of Medical Imaging (Ground Truth):** A critical requirement for clinical adoption is the ability to compare algorithmic results with raw medical images. Future versions of the dashboard must incorporate Curved Multi-Planar Reformation (CMPR) and cross-sectional views of the real Coronary Computed Tomography Angiography (CCTA) scans. This overlay ensures clinicians can validate the software's findings against the anatomical reality.
- **Advanced Dashboard Features and Explainability:** Taking inspiration from robust commercial solutions, the ultimate goal for the user interface is a highly interactive dashboard. Key requested components include a 3D representation of the coronary tree, interactive metrics where doctors can adjust thresholds, and clear explainability of the

algorithmic decisions to avoid "black-box" models. Furthermore, rather than displaying per-point stenosis metrics, the interface should highlight aggregated affected zones (e.g., regions with $\geq 50\%$ stenosis) to allow doctors to quickly triage or inspect specific areas.

4.2.3 Clinical Workflow Integration

The most significant potential of this visualization tool lies in its integration within a broader hospital ecosystem. By unifying automatic segmentation inputs with advanced patient prioritization algorithms, this project serves as a crucial connective element. In a clinical setting, a doctor reviewing a prioritized list of at-risk patients could open a pop-up of this dashboard for a rapid, initial assessment. This quick overview of key metrics and affected zones would allow clinicians to efficiently decide whether a patient requires immediate, detailed analysis using complex diagnostic software or if they can be safely managed otherwise.